

Evolutionary Algorithm for Static Global Robot Localization in 2D Occupancy Maps

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Abstract

This report investigates the use of an Evolutionary Algorithm (EA) for static global localization of a robot in a known two-dimensional occupancy grid. Given a single range observation generated from an unknown robot pose, the task is formulated as a black-box optimization problem in which candidate poses are evaluated according to their compatibility with the observed scan. The project compares a direct scan-matching fitness function with a perception-aware fitness function, studies the sensitivity to the number of range beams, evaluates the effect of EA parameters, and analyses ambiguous cases using top- k localization hypotheses. The final goal is to determine whether a population-based evolutionary search can recover accurate and robust pose estimates, especially in maps where multiple poses may explain similar observations.

1 Introduction

Robot localization is a fundamental problem in autonomous robotics. In order to navigate and interact with an environment, a robot must estimate its pose, usually defined by its position and orientation, with respect to a known map. A particularly challenging case is global localization, where the robot has no reliable initial estimate of its pose. This situation is closely related to the kidnapped robot problem, in which a robot must recover its position after being unexpectedly displaced [6, 2].

Classical localization methods, such as Monte Carlo Localization, address this problem by maintaining a probabilistic belief over the robot pose through a set of weighted particles [3, 6]. These particles are updated over time using motion and observation models. In this project, instead, we focus on a simplified static formulation of the problem: the robot does not move, no motion model is used, and localization is performed from a single range scan in a known 2D occupancy-grid map.

Under these assumptions, localization can be interpreted as an optimization problem. Given a map and an observed scan, the goal is to find the pose whose simulated observation is most consistent with the real one. This formulation makes it possible to approach localization as a direct search problem in the continuous pose space (x, y, θ) , following the common view of optimization as a search over candidate solutions [1].

This project investigates the use of an Evolutionary Algorithm for static global localization. Each individual represents a candidate robot pose, and its quality is evaluated by comparing the scan simulated from that pose with the observed scan. The population is then evolved through selection, crossover, mutation, and elitism, which are standard components of evolutionary algorithms [1, 4]. The population-based nature of Evolutionary Algorithms is particularly relevant in this setting, because several different poses may produce similar observations in symmetric or repetitive environments.

The objective of the project is to evaluate whether an evolutionary search strategy can recover accurate or at least plausible pose hypotheses in controlled 2D maps. Particular attention is given to ambiguity: in some cases, a pose that is far from the ground truth may still be perceptually consistent with the available scan. For this reason, the project does not only evaluate exact localization accuracy, but also studies the presence of scan-equivalent hypotheses.

2 Problem Formulation

The problem considered in this project is a simplified version of the kidnapped robot problem in a known two-dimensional environment [6, 2]. A robot is placed at an unknown pose inside a map and receives a single range observation. The objective is to recover the robot pose using only the known map and the observed scan, without relying on a previous pose estimate or on temporal tracking.

The environment is represented as a 2D occupancy grid map $M : \Omega \subset \mathbb{R}^2 \rightarrow \{0, 1\}$, a standard representation in mobile robotics [5, 6], where $M(x, y) = 0$ denotes free space and $M(x, y) = 1$ denotes an occupied cell. A robot pose is defined as

$$\mathbf{x} = (x, y, \theta) \in \mathcal{X}, \quad (1)$$

where x and y are the robot coordinates in the map and θ is its orientation. Since the robot can only be located in collision-free positions, the feasible pose space $\mathcal{X}_{free}$ contains only poses whose position lies in free space:

$$\mathcal{X}_{free} = \{(x, y, \theta) \in \mathcal{X} \mid M(x, y) = 0\}. \quad (2)$$

In the kidnapped robot setting, the true pose $\mathbf{x}^*$ is unknown to the localization algorithm. It is used only to generate the observed scan and, later, to evaluate the quality of the estimated pose. The observed scan is modeled as

$$\mathbf{z}^{obs} = h(M, \mathbf{x}^*) + \epsilon, \quad (3)$$

where $h(M, \mathbf{x})$ is the range sensor model simulated by ray casting and ϵ represents measurement noise [6]. For a candidate pose $\mathbf{x}$, the expected scan is

$$\mathbf{z}(\mathbf{x}) = h(M, \mathbf{x}). \quad (4)$$

The localization task can therefore be formulated as the search for the candidate pose whose simulated scan is most consistent with the observed scan:

$$\hat{\mathbf{x}} = \arg \min_{\mathbf{x} \in \mathcal{X}_{free}} d(\mathbf{z}^{obs}, \mathbf{z}(\mathbf{x})), \quad (5)$$

where $d(\cdot, \cdot)$ is a scan discrepancy measure. In this project, a natural choice is the mean squared scan error:

$$d(\mathbf{z}^{obs}, \mathbf{z}(\mathbf{x})) = \frac{1}{B} \sum_{i=1}^B (z_i^{obs} - z_i(\mathbf{x}))^2, \quad (6)$$

where B is the number of range beams.

To evaluate the quality of an estimated pose, two main localization errors are considered. The translational error measures the Euclidean distance between the estimated and true positions, while the angular error measures the smallest difference between the estimated and true orientations:

$$e_{pos} = \sqrt{(\hat{x} - x^*)^2 + (\hat{y} - y^*)^2} \quad \text{and} \quad e_{\theta} = \left| \text{wrap}_{[-\pi, \pi]}(\hat{\theta} - \theta^*) \right|. \quad (7)$$

The wrapping operator is required because orientations are periodic. A direct subtraction between two angles may give a large numerical difference even when the two directions are almost equivalent, for example near the discontinuity between $-\pi$ and π . The operator $\text{wrap}_{[-\pi, \pi]}(\cdot)$ maps the angular difference back into the interval $[-\pi, \pi]$, so that e_θ represents the shortest angular distance between the two orientations.

A localization attempt can be considered successful if both errors are below predefined thresholds:

$$e_{pos} \leq \tau_{pos} \quad \text{and} \quad e_\theta \leq \tau_\theta. \quad (8)$$

Intuitively, the problem consists of asking: among all physically possible robot poses in the map, which one would have produced a scan most similar to the scan actually observed? In simple environments this match may be unique, while in symmetric or repetitive maps several distinct poses can produce similar range observations. For this reason, the problem is naturally multimodal and cannot always be reduced to finding a single clearly dominant solution.

3 Implementation

The system was implemented as a lightweight Python pipeline for static global localization in 2D occupancy grid maps. The implementation follows the formulation introduced in Section 2: the robot is assumed to be stationary, the map is known, and the localization algorithm receives only one range scan. Under these assumptions, the task is implemented as a direct search in the continuous pose space, where each candidate solution is evaluated by simulating what the robot would perceive from that pose. A qualitative visual example of the sensing model, scan comparison, and population convergence is provided in Appendix A.1.

3.1 Map and 2D LiDAR Model

The environment is represented as an occupancy grid, where each cell is either free or occupied, following a standard map representation for mobile robot localization [5, 6]. The grid is associated with a metric resolution, so that continuous world coordinates can be converted into discrete map indices. This choice keeps the representation simple and makes collision checking efficient: a pose is considered valid if its position falls inside the map and corresponds to a free cell.

The robot is equipped with a simulated 2D range sensor, modeled as a range-finder observation process [6]. The sensor is parameterized by the number of beams B , the field of view φ , the maximum range $r_{\max}$, and the ray-casting step size Δr . For a robot pose $\mathbf{x} = (x, y, \theta)$, the direction of beam b is computed as

$$\alpha_b = \theta - \frac{\varphi}{2} + b \frac{\varphi}{B-1}, \quad b = 0, \dots, B-1. \quad (9)$$

Each beam is simulated by ray casting. Starting from the robot position, points are sampled along the beam direction until either an occupied cell is reached, the ray exits the map, or the maximum sensor range is exceeded. A point along beam b at distance r is given by

$$p_b(r) = \begin{bmatrix} x + r \cos(\alpha_b) \\ y + r \sin(\alpha_b) \end{bmatrix}. \quad (10)$$

The corresponding range measurement is the first distance at which an obstacle or map boundary is encountered:

$$z_b(\mathbf{x}) = \min \{r \in [0, r_{\max}] \mid M(p_b(r)) = 1 \text{ or } p_b(r) \notin \Omega\}. \quad (11)$$

If no obstacle is detected before $r_{\max}$, the beam returns the maximum range. Optional Gaussian noise can be added to the observed scan and clipped to the valid range interval $[0, r_{\max}]$. In the

experiments, this sensor model is used both to generate the reference observation from the true pose and to evaluate candidate poses during optimization.

This model is intentionally simple. It does not include effects such as beam divergence, reflections, dynamic obstacles, or sensor failures. The goal is not to reproduce a complete physical LiDAR, but to obtain a controlled observation model that is sufficient to test whether an evolutionary search strategy can recover plausible poses from geometric scan information.

3.2 Individual Representation

Each individual in the evolutionary algorithm represents a complete candidate robot pose. This follows the standard EA design principle that candidate solutions must be represented in a form that can be evaluated and modified by variation operators [1]. The genetic part of the individual is a real-valued vector:

$$g_i = (x_i, y_i, \theta_i). \quad (12)$$

The phenotype is the corresponding pose in the occupancy grid map. Since the genotype directly stores the physical pose variables, decoding is straightforward: the values x_i and y_i define the robot position, while θ_i defines its orientation. The only additional constraint is feasibility, since the position must lie inside the map and correspond to a free cell.

In addition to its genetic representation, each individual is associated with the scan that would be perceived from its pose. This scan is obtained by applying the simulated 2D LiDAR model to the map. From this simulated observation, a compact perception descriptor can also be extracted:

$$\phi(\mathbf{z}_i) = [\mu_i, \sigma_i, z_{i,\min}, z_{i,\max}, \rho_i, q_{i,\max}], \quad (13)$$

where μ_i and σ_i are the mean and standard deviation of the ranges, $z_{i,\min}$ and $z_{i,\max}$ are the minimum and maximum measured distances, ρ_i measures the roughness of the scan, and $q_{i,\max}$ is the fraction of beams that reach the maximum sensor range.

The perception descriptor is not part of the genotype and is not modified directly by crossover or mutation. It is instead derived from the phenotype after simulating the sensor. This distinction is important: the evolutionary algorithm still searches in the pose space (x, y, θ) , but each pose can be evaluated not only through beam-wise scan similarity, but also through higher-level information about the structure of the observation.

This extension is useful because two candidate poses can have similar average scan errors while producing scans with different global structure. For example, a pose looking into an open region and a pose located near a wall may differ not only in individual beam values, but also in the distribution, variability, and roughness of the scan. The perception descriptor provides a lightweight way to include this structural information without increasing the dimensionality of the search space. In this sense, it enriches the evaluation model while keeping the genotype simple and computationally cheap to manipulate.

3.3 Initial Population

The initial population is sampled uniformly over the valid pose space, reflecting the population-based generate-and-test structure of evolutionary search [1]. Candidate positions are drawn within the metric bounds of the map and accepted only if they correspond to free cells. Orientations are sampled uniformly in the angular domain:

$$x_i \sim \mathcal{U}(x_{\min}, x_{\max}), \quad y_i \sim \mathcal{U}(y_{\min}, y_{\max}), \quad \theta_i \sim \mathcal{U}(-\pi, \pi). \quad (14)$$

If a sampled position is invalid, it is rejected and resampled. This rejection strategy ensures that the initial population contains only feasible robot poses. The reason for using a uniform initialization is that, in the kidnapped robot problem, no prior information about the robot pose is assumed. The algorithm must therefore begin with a broad exploration of the map.

3.4 Fitness Evaluation

The fitness function evaluates how well a candidate pose explains the observed scan. In evolutionary computation, the evaluation function defines the selection pressure that guides the population through the search space [1, 4]. For each individual $\mathbf{g}_i$, the corresponding pose is used to simulate a predicted scan $\mathbf{z}_i$ through the ray-casting model described above. This predicted scan is then compared with the observed scan $\mathbf{z}^{\text{obs}}$. The basic scan error is implemented as a normalized mean squared error:

$$E_{\text{scan}}(i) = \frac{1}{r_{\text{max}}^2} \frac{1}{B} \sum_{b=1}^B (z_b^{\text{obs}} - z_{i,b})^2. \quad (15)$$

The normalization by r_{max}^2 makes the error less dependent on the absolute sensor range and keeps values comparable across different sensor configurations.

The scan error is transformed into a fitness value to be maximized:

$$F_{\text{scan}}(i) = \frac{1}{1 + E_{\text{scan}}(i)}. \quad (16)$$

With this transformation, lower scan error corresponds to higher fitness. A perfect scan match gives a fitness close to 1, while increasingly poor matches approach 0.

In the perception-aware mode, the scan comparison is extended using the perception descriptor defined in Section 3.2. Rather than comparing only individual beam differences, each candidate pose is also evaluated through the global properties of its perceived scan. Formally, the descriptor $\phi(\mathbf{z})$ introduced in Eq. (13) is computed for both the observed scan $\mathbf{z}^{\text{obs}}$ and the candidate scan $\mathbf{z}_i$, yielding $\phi(\mathbf{z}^{\text{obs}})$ and $\phi(\mathbf{z}_i)$ respectively.

The perception error is then computed as the mean absolute difference between the two normalized descriptor vectors:

$$E_{\text{perc}}(i) = \frac{1}{D} \sum_{j=1}^D |\tilde{\phi}_j(\mathbf{z}^{\text{obs}}) - \tilde{\phi}_j(\mathbf{z}_i)|, \quad (17)$$

where D is the number of descriptor components and $\tilde{\phi}$ denotes the normalized descriptor. The combined error is then

$$E_{\text{combined}}(i) = (1 - w_{\text{perc}}) E_{\text{scan}}(i) + w_{\text{perc}} E_{\text{perc}}(i), \quad (18)$$

where w_{perc} controls the contribution of the perception term. The corresponding perception-aware fitness is

$$F_{\text{perc}}(i) = \frac{1}{1 + E_{\text{combined}}(i)}. \quad (19)$$

The perception descriptor is not an additional gene and is not evolved independently. It is a phenotypic descriptor derived from the scan generated by a candidate pose. This distinction is important: the evolutionary algorithm still searches only in the pose space, while the descriptor provides an alternative way of evaluating the perceptual similarity between poses.

Fitness mode	Beam-wise scan	Perception descriptor	Purpose
Sensor-only	Yes	No	Direct range matching
Perception-aware	Yes	Yes	Add global scan structure

Table 1: Comparison between the two fitness modes.

3.5 Selection

Parent selection is implemented through deterministic tournament selection, a common selection mechanism in evolutionary algorithms [1, 4]. At each selection step, a subset of k individuals is sampled from the current population and the individual with the highest fitness is selected:

$$p = \arg \max_{g_i \in \mathcal{T}} F(g_i), \quad |\mathcal{T}| = k. \quad (20)$$

Tournament selection was chosen for both algorithmic and computational reasons. First, it does not require fitness normalization, which makes it robust to different fitness scales and avoids problems that may arise when raw fitness values are very close to each other. Second, it is simple to implement and only requires comparing a small subset of the population at each selection step. This makes the operator computationally cheap, especially because the expensive part of the algorithm is the scan simulation and fitness evaluation rather than the selection itself.

The tournament size k directly controls the selection pressure. A larger value of k increases the probability of selecting high-fitness individuals and therefore accelerates convergence, but it may also reduce diversity too quickly. A smaller value preserves more exploration, but can slow down convergence. This makes tournament selection a practical choice for this project, since the exploitation–exploration balance can be adjusted with a single intuitive parameter.

3.6 Crossover

Crossover is used to combine information from two parent poses and generate new candidate solutions, following the role of recombination as a variation operator in evolutionary search [1]. Since the genotype is real-valued, binary or permutation-based crossover operators would not be appropriate; real-valued representations require variation operators that match the continuous domain [1]. The implementation therefore uses a blend-like crossover for the position coordinates. Given two parent coordinates a and b , a child coordinate is sampled from an interval that may extend beyond the segment between the parents:

$$c \sim \mathcal{U}(\min(a, b) - \alpha|a - b|, \max(a, b) + \alpha|a - b|), \quad (21)$$

where α controls how far the sampling interval extends beyond the two parent values.

This formulation is useful because it naturally matches the continuous nature of the pose space. If two parents are located in a promising region, the offspring is likely to be generated near that region, supporting local exploitation. At the same time, the extension controlled by α allows the offspring to move slightly outside the segment between the parents, introducing a limited exploratory effect without requiring a completely random restart.

The orientation gene requires special treatment because angles are circular variables. Direct arithmetic averaging can fail near the discontinuity between $-\pi$ and π . For this reason, the implementation blends orientations on the unit circle:

$$\theta_c = \text{atan2}(\alpha \sin \theta_1 + (1 - \alpha) \sin \theta_2, \alpha \cos \theta_1 + (1 - \alpha) \cos \theta_2). \quad (22)$$

This circular formulation avoids artificial jumps caused by angular wrap-around and produces a meaningful intermediate orientation between the two parents. After crossover, the resulting pose is checked against the occupancy grid. If the child position is invalid, the operator retries for a limited number of attempts; if no valid child is found, the algorithm falls back to copying the fitter parent. This validity check prevents infeasible individuals from propagating through the population and avoids wasting evaluations on poses that cannot correspond to a real robot configuration.

3.7 Mutation

Mutation introduces random local changes in the candidate pose. Gaussian perturbation is a standard mutation mechanism for real-valued evolutionary search [1]. With probability p_m , Gaussian noise is added independently to the position and orientation components:

$$x' = x + \eta_x, \quad y' = y + \eta_y, \quad \theta' = \text{wrap}_{[-\pi, \pi]}(\theta + \eta_\theta), \quad (23)$$

with

$$\eta_x, \eta_y \sim \mathcal{N}(0, \sigma_{xy}^2), \quad \eta_\theta \sim \mathcal{N}(0, \sigma_\theta^2). \quad (24)$$

Gaussian mutation was chosen because it is well suited to continuous pose optimization. Small perturbations are more likely than large ones, so the operator naturally performs local refinement around promising poses. At the same time, the non-zero tails of the Gaussian distribution still allow occasional larger moves, which can help the population escape poor local regions.

Separate step sizes are used for translation and rotation because position and orientation have different physical meanings and different numerical scales. A displacement of a few centimeters and a rotation of a few radians cannot be treated as equivalent perturbations. The parameters σ_{xy} and σ_θ therefore make it possible to control spatial and angular exploration independently.

After mutation, the orientation is wrapped back into $[-\pi, \pi]$, while the position is checked against the map. If the mutated pose lies outside the map or inside an occupied cell, the mutation is retried for a limited number of attempts. If no valid mutation is obtained, the original pose is preserved. This strategy is computationally convenient because it keeps the population feasible and avoids spending ray-casting evaluations on invalid robot configurations.

3.8 Survivor Selection and Termination

The algorithm follows a generational scheme with elitism, a common survivor-selection strategy used to preserve high-quality solutions across generations [1]. At each generation, the best e individuals are copied directly to the next population:

$$\mathcal{E}_t = \text{Top}_e(\mathcal{P}_t), \quad (25)$$

where $\mathcal{P}_t$ is the population at generation t . The remaining positions are filled by offspring generated through selection, crossover, mutation, and evaluation. Elitism is used to prevent the best solutions found so far from being lost due to stochastic variation.

The run terminates after a fixed number of generations. During evolution, both the best fitness and the mean population fitness are recorded. The best fitness indicates the quality of the current best pose hypothesis, while the mean fitness gives information about the overall convergence of the population.

3.9 Top- k Hypothesis Extraction

In ambiguous maps, the single best individual is not always sufficient to describe the result of localization. Different poses may produce very similar scans, especially in corridors or repeated structures. For this reason, the implementation also extracts a set of top- k diverse hypotheses from the final population.

Individuals are first sorted by fitness. Then, candidates are added to the top- k set only if they are sufficiently different from the poses already selected. The distance between two poses combines translational and angular differences:

$$d_{\text{pose}}(\mathbf{x}_a, \mathbf{x}_b) = \sqrt{(x_a - x_b)^2 + (y_a - y_b)^2} + \lambda_\theta \left| \text{wrap}_{[-\pi, \pi]}(\theta_a - \theta_b) \right|. \quad (26)$$

This procedure favors high-fitness hypotheses while avoiding multiple nearly identical copies of the same solution. It is particularly useful for analyzing whether a failure is due to poor convergence of the evolutionary algorithm or to genuine perceptual ambiguity in the map.

Component	Implementation	Motivation
Representation	Real-valued pose (x, y, θ)	Matches the continuous nature of the localization space.
Perception	Scan-derived descriptor $\phi(\mathbf{z})$	Adds structural information without increasing the search-space dimensionality.
Initial population	Uniform sampling of valid poses	Reflects the absence of prior pose information in global localization.
Parent selection	Tournament selection	Computationally cheap and independent of absolute fitness scaling.
Crossover	Blend crossover and circular angle interpolation	Suitable for real-valued genes and avoids angular discontinuities.
Mutation	Gaussian perturbation of position and orientation	Supports local refinement while still allowing occasional larger moves.
Validity handling	Rejection or resampling of invalid poses	Prevents evaluations of infeasible robot configurations.
Survivor strategy	Generational replacement with elitism	Preserves the best solutions found during the search.
Termination	Fixed number of generations	Provides a simple and reproducible computational budget.

Table 2: Main evolutionary components used in the implementation.

4 Experimental Setup

The experimental evaluation is designed to assess the behaviour of the proposed evolutionary localization method under different map structures, sensor configurations, and algorithmic settings, following the need to evaluate stochastic EAs over multiple independent runs [1]. Since the problem is stochastic, each experiment is repeated over multiple random seeds, i.e a list of seeds ($\text{seeds} \in [0, 10]$). For each run, the algorithm receives the map and a single observed range scan generated from a known true pose, while the true pose itself is used only for evaluation.

The experiments are carried out on five synthetic 2D occupancy-grid maps, shown in Figure 1. The maps are designed to cover different levels of structural complexity, from simple open rooms to environments with repeated or symmetric structures. This allows the localization method to be evaluated both in relatively easy scenarios and in more ambiguous cases.

The base configuration used across all experiments is reported in Tables 3 and 4. Individual experiments deviate from this baseline only for the parameters under study.

Param.	Description	Value
N	Population size	100
T	Number of generations	40
k	Tournament size	3
p_c	Crossover rate	0.9
p_m	Mutation rate	0.8
σ_{xy}	Spatial mutation step	0.35 m
σ_θ	Angular mutation step	0.25 rad
e	Elite size	3
w_{perc}	Perception weight	0.3

Table 3: Base EA configuration.

Param.	Description	Value
B	Number of beams	72
φ	Field of view	360°
r_{max}	Maximum range	8.0 m
Δr	Ray-casting step	0.05 m
σ_ϵ	Measurement noise std	0.0 m

Table 4: Base sensor configuration. Noise is zero to isolate the fitness landscape from stochastic observation effects.

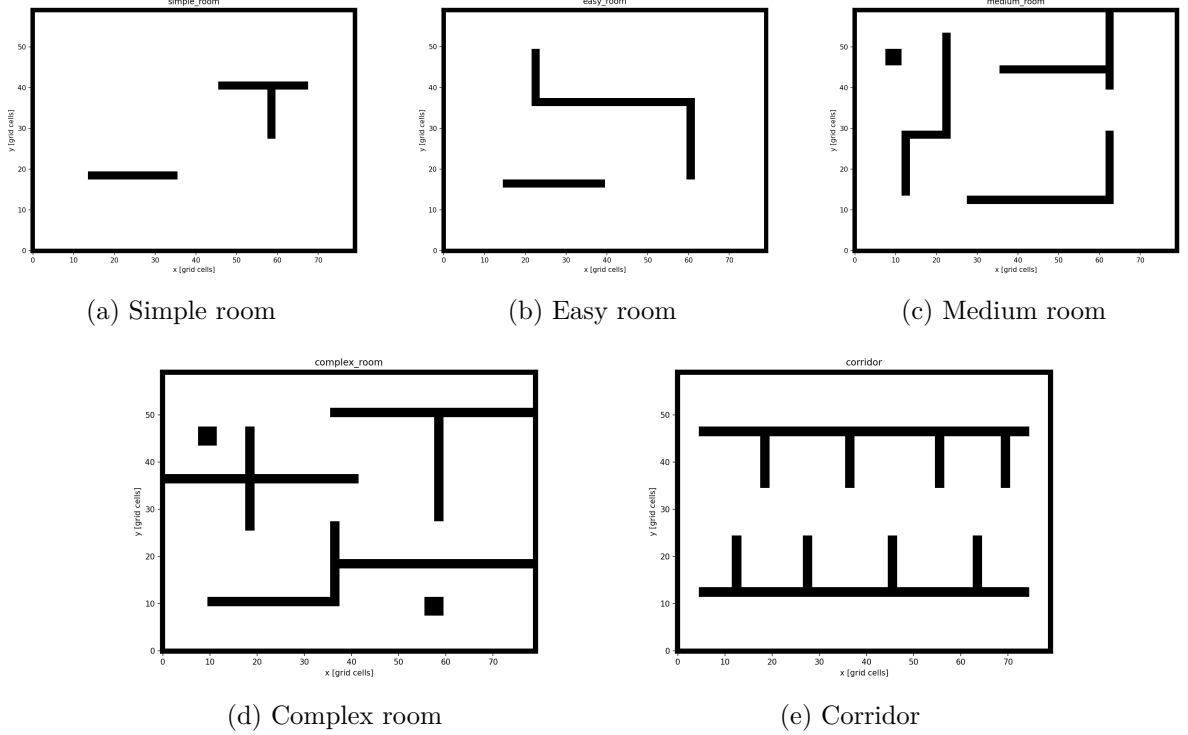


Figure 1: Synthetic occupancy-grid maps used in the experimental evaluation. Black cells represent occupied space, while white cells represent free space. The maps progressively introduce more complex geometry, repeated structures, and potential ambiguities for scan-based localization.

4.1 Evaluation Metrics

For each run, the final estimated pose is compared with the true pose using the position and angular errors defined in Section 2. A run is considered successful if both errors are below predefined thresholds:

$$S = \mathbb{I}[e_{pos} \leq \tau_{pos} \wedge e_{\theta} \leq \tau_{\theta}], \quad (27)$$

where $S = 1$ indicates a successful localization and $S = 0$ indicates failure.

Over multiple runs, the success rate is computed as

$$R_{succ} = \frac{1}{N} \sum_{j=1}^N S_j, \quad (28)$$

where N is the number of independent runs. In addition to success rate, the experiments report the mean and standard deviation of the position error, angular error, final best fitness, and runtime. These quantities allow us to evaluate not only whether the algorithm converges to the correct pose, but also how stable and computationally expensive the convergence is.

In ambiguous environments, the single best individual may not be sufficient to describe the behaviour of the algorithm. For this reason, some experiments also consider a top- k success criterion, which is consistent with the population-based view of EAs as methods that can preserve multiple candidate solutions in multimodal search spaces [1]. Given a set of diverse high-fitness hypotheses $\mathcal{H}_k$, the run is successful if at least one candidate pose satisfies the localization thresholds:

$$S_{top-k} = \mathbb{I}[\exists \mathbf{x}_i \in \mathcal{H}_k : e_{pos}(\mathbf{x}_i) \leq \tau_{pos} \wedge e_{\theta}(\mathbf{x}_i) \leq \tau_{\theta}]. \quad (29)$$

This metric is useful because, in multimodal maps, the population may contain the correct pose even when the single best individual corresponds to another plausible hypothesis.

4.2 Experiment 1: Perception Ablation

The first experiment evaluates the effect of the perception-aware fitness term. The same localization tasks are solved using two fitness modes:

- A *sensor-only fitness*, based directly on beam-wise scan comparison,
- A *perception-aware fitness*, which also includes the descriptor extracted from the simulated scan.

The purpose of this experiment is to understand whether adding global scan descriptors improves the robustness of the localization process. The sensor-only fitness is expected to be more direct and precise when the scan match is informative, while the perception-aware version may provide additional guidance in cases where raw beam-wise differences are noisy or partially ambiguous. For each map and seed, both methods are run under the same EA configuration. The comparison focuses on final localization error, success rate, and final best fitness.

4.3 Experiment 2: Benchmark Across Maps

The second experiment evaluates the method across all available map types. The goal is to study how the difficulty of the localization problem changes with the geometry of the environment.

Simple maps are expected to produce smoother and less ambiguous fitness landscapes, making localization easier. More complex or symmetric maps are expected to be more challenging because several different poses may produce similar range scans. This experiment therefore measures the robustness of the method with respect to environmental structure.

For each map, the algorithm is run from multiple random seeds and the results are summarized in terms of position error, angular error, success rate, runtime, and fitness. This provides a general benchmark of the localization pipeline.

4.4 Experiment 3: Beam Sensitivity

The third experiment studies the effect of sensor resolution by varying the number of LiDAR beams B . A larger number of beams provides a richer observation, but it also increases the computational cost of each fitness evaluation, since every candidate pose requires a complete ray-casting simulation.

The main objective is to evaluate the trade-off between localization accuracy and computational cost. With too few beams, the scan may not contain enough information to distinguish similar poses. With more beams, the observation becomes more informative, but each generation becomes more expensive.

This experiment compares different values of B while keeping the remaining EA parameters fixed. The expected outcome is that accuracy improves as the number of beams increases, but with diminishing returns after a certain point.

4.5 Experiment 4: Parameter Tuning

The fourth experiment investigates the sensitivity of the evolutionary algorithm to its main parameters. In particular, different configurations of population size, number of generations, mutation strength, crossover behaviour, and tournament size are tested.

The purpose of this experiment is not to perform exhaustive hyperparameter optimization, but to identify which settings provide a good balance between exploration, convergence quality, and runtime; parameter tuning is itself a central issue in evolutionary algorithm design [1]. Larger populations and more generations should generally improve the probability of finding good poses, but they also increase computational cost. Similarly, stronger mutation may help exploration in

ambiguous maps, while smaller mutation may improve local refinement once a promising region has been found.

Each configuration is evaluated over multiple seeds, and the comparison is based on success rate, localization errors, final fitness, and runtime.

4.6 Experiment 5: Ambiguity and Symmetry Analysis

The final experiment focuses on ambiguous localization cases. In symmetric or repetitive maps, a wrong estimated pose may still generate a scan that is very similar to the observed one. In such cases, failure according to the ground-truth pose may not necessarily indicate poor optimization, but rather genuine perceptual ambiguity.

To study this effect, the algorithm is evaluated on maps and poses where scan similarity can occur at multiple spatial locations. The final population is analyzed using diverse top- k hypotheses, and candidate poses are compared not only with the true pose, but also with other scan-equivalent or near-equivalent poses.

The objective is to distinguish between two types of failures. The first is an optimization failure, where the algorithm does not find any pose compatible with the observation. The second is an ambiguity-driven failure, where the algorithm finds a plausible pose that is different from the ground truth but produces a very similar scan. This distinction is important because, in static localization from a single observation, some ambiguities cannot be resolved without additional information such as motion, temporal filtering, or a richer sensor model.

5 Results

This section reports the main experimental results obtained from the evolutionary localization pipeline. The focus is not only on the final success rate, but also on the interpretation of the errors and on what the results reveal about the suitability of Evolutionary Algorithms for static global localization, especially given their population-based search behaviour [1].

Across the experiments, the median localization errors are often much smaller than the corresponding mean errors. This indicates that, in many runs, the algorithm converges very close to the true pose, while a smaller number of failures produces large position or orientation errors. For this reason, both mean and median values are reported when possible.

5.1 Effect of the Perception-Aware Fitness

The first experiment compares the sensor-only fitness with the perception-aware fitness. The results are summarized in Table 5.

Method	Runs	Succ.	Mean e_{pos}	Med. e_{pos}	Mean e_{θ}	Med. e_{θ}	Time [s]
Sensor-only	250	0.564	1.798	0.068	1.139	0.025	11.695
Perception-aware	250	0.668	1.473	0.053	0.847	0.019	11.881

Table 5: Comparison between sensor-only and perception-aware fitness.

The perception-aware fitness improves the success rate from 56.4% to 66.8%, corresponding to an absolute gain of 10.4 percentage points. It also reduces the mean position error from 1.798 to 1.473, and the mean angular error from 1.139 to 0.847. The median errors are low for both methods, but they also improve with the perception-aware version.

This suggests that the additional perception descriptor $\phi(\cdot)$ is useful because it provides global information about the structure of the scan, rather than relying only on beam-wise range differences. Importantly, this improvement is obtained with almost no additional runtime cost:

the average runtime increases only from 11.695 s to 11.881 s. Therefore, the perception-aware fitness improves robustness without significantly increasing the computational burden.

At the same time, the improvement is not absolute. The success rate remains below 70%, showing that the descriptor helps the search but does not fully solve ambiguity or local convergence issues. This is coherent with the nature of the problem: if two poses produce genuinely similar observations, a richer scan descriptor may still be insufficient to distinguish them.

5.2 Performance Across Maps

The second experiment evaluates the perception-aware version across the different maps. The results are reported in Table 6.

Map	Runs	Succ.	Mean e_{pos}	Med. e_{pos}	Mean e_{θ}	Med. e_{θ}	Time [s]
Simple room	50	0.760	1.533	0.036	0.755	0.009	15.491
Easy room	50	0.660	1.763	0.055	1.052	0.022	12.235
Medium room	50	0.680	1.305	0.057	0.682	0.019	11.010
Complex room	50	0.540	1.527	0.123	0.801	0.044	9.310
Corridor	50	0.700	1.235	0.044	0.948	0.016	11.360

Table 6: Localization performance across maps using the perception-aware fitness.

The simple room gives the best success rate, 76.0%, while the complex room is the most difficult scenario, with a success rate of 54.0%. This result is expected: as the map becomes more structured and contains more obstacles or repeated patterns, the fitness landscape becomes harder to optimize and may contain more local optima.

A relevant observation is that the median position error remains very low in all maps, ranging from 0.036 to 0.123. This means that when the algorithm localizes correctly, it usually does so with high precision. The larger mean errors are mainly caused by runs in which the algorithm converges to a wrong but still plausible region of the map.

The corridor is particularly interesting. Although it reaches a relatively good success rate of 70.0%, its mean angular error remains high. This suggests that corridor-like environments can be geometrically ambiguous: different poses along the corridor may produce similar scans, especially when the sensor observation is not enough to uniquely determine the position and orientation.

5.3 Effect of the Number of LiDAR Beams

The beam sensitivity experiment evaluates how the number of LiDAR beams affects localization performance and runtime. Table 7 reports the success rate for each tested beam configuration.

Map	18 beams	36 beams	72 beams	108 beams	144 beams
Simple room	0.72	0.72	0.76	0.70	0.68
Complex room	0.40	0.62	0.54	0.54	0.50
Corridor	0.40	0.68	0.70	0.72	0.60

Table 7: Success rate for different numbers of LiDAR beams.

The results show that increasing the number of beams does not always monotonically improve localization. In the simple room, the best success rate is obtained with 72 beams, while increasing the resolution further to 108 or 144 beams slightly reduces performance. In the complex room, the best result is obtained with 36 beams, and in the corridor the best result is obtained with 108 beams.

This suggests that more sensor information is not automatically better for the evolutionary search. More beams make the scan comparison more detailed, but they also make each fitness

evaluation more expensive and may make the fitness landscape sharper. A sharper landscape can help local refinement once the population is close to the correct pose, but it can also make global exploration harder.

The computational cost increases almost linearly with the number of beams. For example, in the simple room, the average runtime increases from 4.342 s with 18 beams to 30.378 s with 144 beams. In the corridor, the runtime increases from 3.296 s to 21.809 s over the same beam range. Therefore, intermediate beam resolutions appear to provide the best trade-off between accuracy and computational cost.

5.4 Effect of EA Parameters

The fourth experiment studies the effect of different EA parameter configurations. The most relevant results are shown in Table 8.

Config.	Runs	Succ.	Mean e_{pos}	Med. e_{pos}	Mean e_θ	Med. e_θ	Time [s]
Base	90	0.600	1.602	0.063	0.897	0.033	10.537
Pop. 50	90	0.411	2.077	0.587	1.498	1.530	5.123
Pop. 150	90	0.733	1.086	0.047	0.656	0.022	16.012
Gen. 80	90	0.611	1.590	0.047	0.888	0.018	20.923
$\sigma_{xy} = 0.15$	90	0.611	1.623	0.042	0.961	0.017	10.740
$\sigma_{xy} = 0.50$	90	0.589	1.739	0.095	0.975	0.034	10.320
Tournament 2	90	0.589	1.659	0.100	0.985	0.035	10.552
Tournament 5	90	0.600	1.547	0.037	0.991	0.009	10.515

Table 8: Effect of EA parameter configurations.

The population size has the strongest effect among the tested parameters. Reducing the population to 50 individuals decreases the success rate to 41.1%, while increasing it to 150 individuals improves the success rate to 73.3%. This is one of the clearest results of the evaluation: for this localization task, population diversity is crucial. A larger population increases the probability of covering multiple promising regions of the pose space, which is especially important in global localization where no prior pose estimate is available.

Increasing the number of generations to 80 does not produce the same improvement. The success rate only increases slightly from 60.0% to 61.1%, while the runtime almost doubles, from 10.537 s to 20.923 s. This indicates that simply running longer is less effective than maintaining a larger and more diverse population. In other words, if the population does not contain promising hypotheses early enough, additional generations may only refine incorrect regions.

The mutation experiments show a more nuanced effect. A smaller spatial mutation step, $\sigma_{xy} = 0.15$, gives a slight improvement over the base configuration, while a larger step, $\sigma_{xy} = 0.50$, slightly decreases performance. This suggests that the algorithm benefits from moderate local refinement rather than overly aggressive perturbations. However, the effect is smaller than the effect of population size.

The tournament size has limited impact on the overall success rate. Tournament size 5 improves the median errors but does not improve the success rate compared with the base configuration. This suggests that stronger selection pressure can refine good solutions more quickly, but may not solve the main difficulty of the task, which is discovering the correct basin of attraction in the first place.

5.5 Ambiguity and Symmetry Analysis

The ambiguity analysis is used to better interpret localization failures in maps where different poses may generate very similar range observations. In these environments, an estimated pose can be far from the ground truth in Euclidean distance while still producing a scan that is

almost indistinguishable from the observed one. Therefore, evaluating the algorithm only with respect to the exact ground-truth pose may underestimate the quality of the solution found by the EA. It is important to notice that, for this experiment, the poses chosen in the corridor map are significantly ambiguous places than the previous experiments, and for this reason the results differ so much.

To show this, Figures 2a and 2b provide a visual explanation of the ambiguity metrics. In both maps, scan-equivalent regions are not concentrated only around the true pose, but also appear in other parts of the environment. This means that different poses can produce very similar range observations. In these figures the blue star indicates the true pose, while circular samples mark poses whose simulated scan is sufficiently similar to the observed scan. The color scale reports the best scan similarity over orientation.

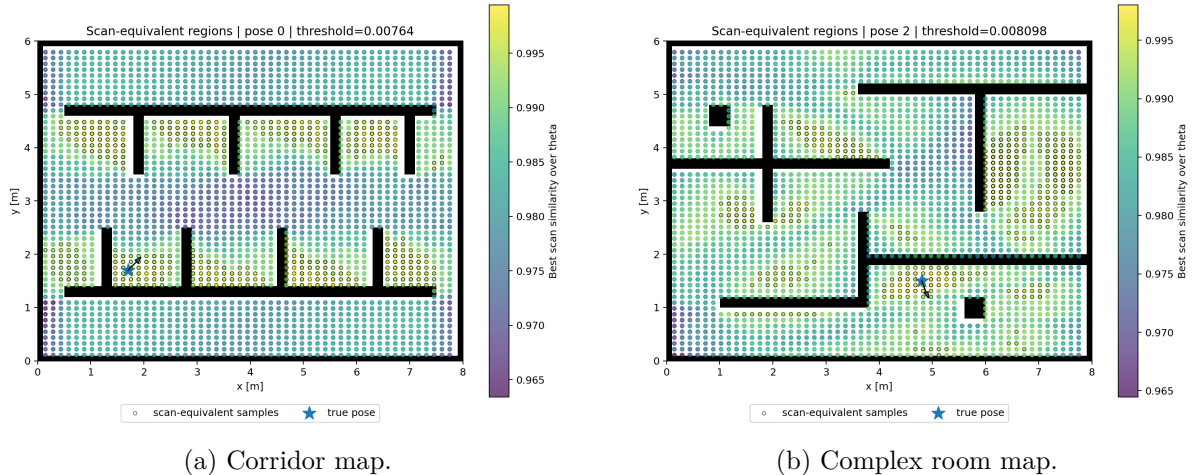


Figure 2: Scan-equivalent regions for representative poses in the corridor and complex room maps. The figures show that multiple spatially distinct regions can generate observations similar to the true scan, making exact pose recovery ambiguous from a single static observation.

The EA outputs in Figure 3 show that the algorithm can converge to one of these scan-equivalent regions. Therefore, a large Euclidean pose error does not necessarily imply that the optimization failed. In these cases, the estimated pose is wrong with respect to the ground truth, but still consistent with the available sensor information.

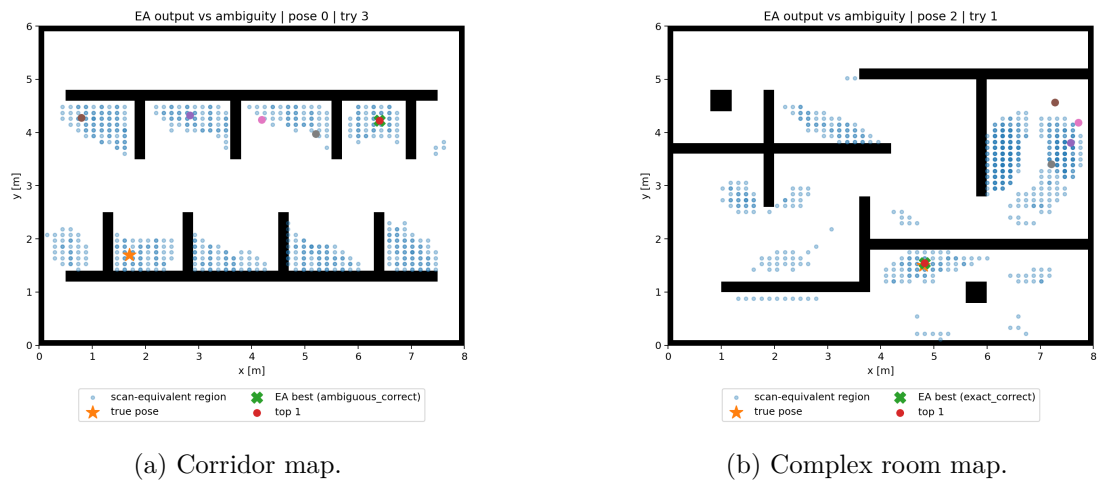


Figure 3: EA output compared with the scan-equivalent regions. In the corridor example, the EA converges to a scan-equivalent region that is far from the ground-truth pose, showing an ambiguity-driven failure. In the complex room example, the best EA solution corresponds to the exact pose, while the other top hypotheses lie in perceptually valid regions.

In these figures, the orange star indicates the true pose, the light-blue samples indicate scan-equivalent poses, and the green cross marks the best solution found by the EA. The additional circular markers represent the top-5 hypotheses ranked by fitness: red is top-1, purple is top-2, pink is top-3, gray is top-4, and brown is top-5.

For this reason, we distinguish between three cases. An *exact success* occurs when the estimated pose satisfies the standard position and angular thresholds with respect to the true pose. A *scan-equivalent success* occurs when the estimated pose is not necessarily close to the true pose, but produces a scan sufficiently similar to the observed scan. Finally, a *true failure* occurs when the estimated pose is neither exact nor scan-equivalent.

The difference between scan-equivalent success and exact success is called ambiguity gain:

$$G_{\text{amb}} = R_{\text{scan-eq}} - R_{\text{exact}}. \quad (30)$$

This quantity measures the fraction of runs that would be considered failures using only the ground-truth pose error, but that are actually compatible with the available observation. A high ambiguity gain therefore indicates that many apparent errors are caused by perceptual ambiguity rather than by poor optimization.

Similarly, the true failure rate is defined as:

$$R_{\text{true-fail}} = 1 - R_{\text{scan-eq}}. \quad (31)$$

This measures how often the algorithm fails to find both the true pose and any scan-equivalent alternative.

Table 9 reports the ambiguity metrics for the two maps used in this analysis.

Metric	Corridor	Complex room
Runs	50	50
Exact success	0.40	0.54
Scan-equivalent success	1.00	0.98
Ambiguity gain	0.60	0.44
True failure	0.00	0.02
Median scan error	0.0005	0.0010
Median position error	1.7978	0.1229
Median angular error	0.0000	0.0440

Table 9: Ambiguity analysis by map.

The corridor map is the clearest example of perceptual ambiguity. The exact success rate is only 40.0%, meaning that less than half of the runs converge to the ground-truth pose according to the standard localization thresholds. However, the scan-equivalent success rate is 100.0%, meaning that every run finds a pose that produces a scan compatible with the observed one. The ambiguity gain of 0.60 is therefore substantial: 60.0% of the runs are apparent failures only because the estimated pose is different from the ground truth, not because it explains the observation poorly.

The complex room shows the same effect, although less strongly. The exact success rate is 54.0%, while the scan-equivalent success rate reaches 98.0%. This produces an ambiguity gain of 0.44, showing that almost half of the runs are better interpreted as scan-equivalent solutions rather than true failures. Only 2.0% of the runs are true failures, meaning that the EA almost always finds either the correct pose or a perceptually valid alternative.

The error statistics clarify why this distinction is important. In the corridor, the median position error is 1.7978, which would normally suggest poor localization. However, the median scan error is only 0.0005, showing that these estimates reproduce the observed scan extremely well. In other words, the algorithm is not simply returning random wrong poses: it is often

converging to poses that are far from the true position but almost equivalent from the point of view of the available sensor observation.

A second analysis groups the runs by failure type. This helps separate exact correct localizations, ambiguous but scan-consistent estimates, and genuine failures. Table 10 reports both the success indicators and the median errors for each group.

Metric	Ambiguous correct	Exact correct	True failure
Runs	52	47	1
Exact success	0.00	1.00	0.00
Scan-equivalent success	1.00	1.00	0.00
True failure	0.00	0.00	1.00
Median scan error	0.0015	0.0001	0.0129
Median position error	3.5323	0.0517	3.2035
Median angular error	0.0000	0.0180	1.2750

Table 10: Ambiguity analysis by failure type.

The failure-type analysis confirms that most non-exact results are not optimization failures. The ambiguous-correct group contains 52 runs: none of them is an exact success, but all of them are scan-equivalent successes. Their median position error is high, 3.5323, but their median scan error is very low, 0.0015. This means that these poses are spatially far from the ground truth, but they are still very good explanations of the observed scan.

The exact-correct group behaves as expected: it has both low pose error and extremely low scan error. The true-failure group, instead, contains only one run. It has a higher scan error and a large angular error, which indicates that the algorithm did not find a pose consistent with the observation.

Overall, this experiment shows that many localization errors in ambiguous maps are not caused by the EA failing to optimize the objective. Instead, they arise because the observation itself is insufficient to uniquely identify the true pose. In a static kidnapped robot setting with a single range scan, different locations can be perceptually equivalent. The population-based nature of the EA is useful in this context because it can reveal multiple plausible hypotheses, but resolving these ambiguities would require additional information, such as temporal motion, a motion model, or richer sensor observations.

5.6 Summary of Main Findings

The experimental results show three main trends. First, the EA can produce highly accurate pose estimates when the observation is sufficiently informative, as indicated by the low median localization errors across several experiments. Second, population diversity plays a central role: increasing the population size improves performance more clearly than simply increasing the number of generations. Third, many apparent localization failures in ambiguous maps are better interpreted as scan-equivalent solutions rather than pure optimization failures.

These findings motivate the broader discussion in Section 6, where the results are interpreted in terms of fitness design, computational cost, parameter sensitivity, ambiguity, and limitations of the current static formulation.

6 Discussion

The results show that Evolutionary Algorithms can be effectively used as a global search method for static robot localization in 2D occupancy-grid maps [1, 4]. The main strength of the approach is its population-based nature: instead of committing immediately to a single pose hypothesis,

the algorithm explores several regions of the pose space in parallel. This is particularly relevant for kidnapped robot recovery, where no reliable initial pose estimate is available [6, 2].

At the same time, the experiments also show that the problem is not only an optimization problem, but also an observability problem. In some maps, especially those with repeated or symmetric structures, several poses can produce almost identical range observations. In these cases, even a successful optimization process may return a pose that is different from the ground truth but still perceptually valid.

6.1 Perception-Aware Ablation

The perception-aware fitness improves the robustness of the localization process by extending the evaluation beyond direct beam-wise scan comparison. Its main contribution is that it adds global information about the structure of the scan, such as range distribution and variability, without changing the representation of the individual. The evolutionary algorithm still searches in the same pose space (x, y, θ) , but each pose is evaluated using a richer description of what the robot would perceive from that location.

This is useful because two poses may have similar local scan errors while differing in the global structure of their observations. The perception descriptor helps reduce this effect by providing additional information to guide selection. From an implementation point of view, this is also convenient because the descriptor is computed after ray casting and does not increase the dimensionality of the genotype or complicate the variation operators.

However, the descriptor used here (mean, standard deviation, minimum, maximum, roughness, and fraction of beams at maximum range) is relatively coarse, and some of its components are likely correlated. Two poses with similar range distributions may still differ structurally in ways that these six values fail to capture. A richer descriptor, or a learned embedding of the scan, could provide stronger guidance. More fundamentally, if two poses generate genuinely similar scans, their descriptors will also be similar regardless of how detailed they are: the limit is the information content of the single static observation, not only the choice of descriptor.

6.2 Beam Information and Computational Cost

The beam sensitivity experiment highlights a clear trade-off between observation richness and computational cost. A low number of beams reduces the cost of each fitness evaluation, because fewer rays have to be simulated for every candidate pose. However, sparse scans may not contain enough geometric information to distinguish between similar poses, especially in maps with repeated structures.

The effect is not monotonic, and this deserves interpretation beyond the numerical results. With few beams, the fitness landscape is smooth but poorly informative: many distant poses appear similar because the sparse scan cannot resolve local geometry. As beam count increases, the landscape becomes sharper. This sharpness aids local refinement once the population is already near the correct region, but it can make global exploration harder, because the basins of attraction around correct poses become narrower and easier to miss during the initial uniform sampling. In this sense, beam resolution acts as an implicit regularization parameter on the fitness landscape, and intermediate values may be preferable not only for computational reasons but for algorithmic ones.

The computational cost scales roughly linearly with the number of beams, since ray casting is performed for every individual at every generation. For example, in the simple room, average runtime increases from approximately 4s with 18 beams to 30s with 144 beams. Intermediate resolutions therefore offer the best practical compromise between accuracy and runtime.

6.3 Sensitivity to EA Parameters

Population size is the most impactful parameter tested. This is consistent with the global nature of the localisation problem: without a prior pose estimate, the algorithm must cover several distant regions of the map simultaneously. A larger population increases the probability that some individuals land in the correct basin of attraction early in the search, which is the prerequisite for successful convergence.

Increasing the number of generations has a different effect. Additional generations mainly allow the algorithm to refine the hypotheses already present in the population. This is useful only if the correct region has already been discovered. If the population has converged to a plausible but incorrect area, more generations will refine that wrong hypothesis rather than recover the true pose. This result was somewhat unexpected: increasing the number of generations was anticipated to have a comparable effect to increasing population size, but the experiments show clearly that early coverage of the pose space matters more than extended refinement.

Mutation step size controls the balance between exploration and local refinement. Small perturbations support precise improvement near a promising pose, while larger perturbations increase the probability of escaping local optima at the risk of disrupting good solutions. Tournament size controls selection pressure: larger tournaments favour faster exploitation of high-fitness individuals, while smaller tournaments preserve more diversity. In the tested configurations, both parameters affect performance, but less strongly than population size.

Overall, the tuning results suggest that early exploration is critical. For static global localisation, maintaining diversity in the initial phases of the search is more important than increasing exploitation pressure or extending the run length.

6.4 Ambiguity, Multimodality, and Top- k Hypotheses

The ambiguity analysis is the key result for understanding the behaviour of the algorithm. In the corridor and complex room maps, the exact success rate is significantly lower than the scan-equivalent success rate. This means that many estimates classified as wrong by the ground-truth pose metric are actually consistent with the observed scan.

This distinction is important because static localization from a single range scan is not always identifiable. If different poses produce nearly identical observations, the algorithm has no additional information to decide which one corresponds to the true physical pose. In such cases, a large Euclidean pose error does not necessarily indicate a failure of the optimization process. It may instead indicate that the observation admits multiple valid explanations.

The top- k hypothesis extraction is useful precisely in this situation. Considering only the best individual can hide the structure of the final population. By looking at multiple high-fitness and diverse hypotheses, it becomes possible to see whether the EA has preserved alternative scan-consistent poses. This better reflects the nature of the problem, because in ambiguous environments the desired output may be a set of plausible hypotheses rather than a single pose.

From this perspective, the population-based nature of EAs is a strength [1]. The algorithm is naturally able to maintain multiple candidate solutions during the search. However, maintaining multiple hypotheses is different from resolving them. To choose between scan-equivalent poses, additional information would be required, such as motion over time, a probabilistic belief update, or richer sensory input.

6.5 Limitations

The current implementation has several limitations. First, the robot is static and no motion model is used. This makes the problem suitable for direct optimization, but it also removes temporal information that would be essential in a complete localization system.

Second, the experiments are performed on synthetic occupancy-grid maps. This provides a controlled and reproducible evaluation, but does not include the full complexity of real

environments, such as map noise, dynamic obstacles, partial observability, or irregular geometry.

Third, the sensor model is based on ideal ray casting. This is useful for testing the algorithmic idea, but it does not model all the effects of real LiDAR sensors, such as missing returns, outliers, reflectance effects, beam divergence, or calibration errors.

Fourth, the method uses only a single scan. This is the main reason why ambiguity becomes important: some poses cannot be distinguished from one observation alone. A sequence of scans would provide additional constraints and could reduce the number of scan-equivalent hypotheses.

Finally, parameter tuning is lightweight rather than exhaustive, and the method has not yet been tested on real robot data. More systematic tuning procedures could be considered in future work, since EA performance can be highly dependent on parameter choices [1]. Therefore, the current results should be interpreted as a proof of concept for EA-based static global localization, not as a complete replacement for classical probabilistic localization methods.

7 Conclusion

This project investigated the use of an Evolutionary Algorithm for static global robot localization in known 2D occupancy-grid maps. The localization problem was formulated as a direct search in the continuous pose space, where each candidate pose is evaluated by comparing a simulated 2D LiDAR scan with a single observed scan. The goal was not to replace full probabilistic localization methods [6, 3], but to study whether a population-based optimizer can recover accurate or plausible pose hypotheses in a simplified kidnapped robot setting.

The experimental results show that the proposed approach can achieve accurate localization when the observation is sufficiently informative. In successful runs, the final pose is often very close to the ground truth, as indicated by the low median position and angular errors. The perception-aware fitness improves robustness by adding global scan-structure information to the beam-wise scan comparison, while keeping the individual representation simple and computationally lightweight.

The results also highlight the importance of population diversity. Increasing the population size has a stronger effect than simply increasing the number of generations, suggesting that broad exploration of the pose space is crucial in global localization. The beam sensitivity experiment shows a trade-off between observation richness and runtime: more beams provide more information, but they also increase the cost of each fitness evaluation and do not always lead to proportional improvements.

The most important finding concerns ambiguity. In symmetric or repetitive maps, several poses can produce very similar range observations. The ambiguity analysis shows that many apparent localization failures are actually scan-equivalent solutions: they are far from the ground truth but still explain the observed scan well. This confirms that, when only a single static scan is available, exact pose recovery is not always identifiable from the observation alone.

Overall, the project shows that Evolutionary Algorithms are a promising tool for static global localization, especially as a global hypothesis generation method. Their population-based structure makes them suitable for multimodal search spaces and allows the algorithm to preserve multiple plausible pose hypotheses [1]. However, the method is limited by the simplified sensor model, synthetic maps, lack of motion information, and absence of real robot data. Future work could extend the approach by introducing temporal information, combining the EA with probabilistic tracking methods such as particle filters, evaluating the method on real LiDAR data, and improving the treatment of ambiguous hypotheses.

A Additional Material

A.1 Visual Case Study

This appendix provides a visual case study of a representative localization run. The goal is to illustrate the main components of the problem: the simulated 2D LiDAR observation, the scan-matching objective, and the convergence of the evolutionary population over time. These figures are intended as qualitative support for the experimental results discussed in the main text.

A.1.1 Simulated 2D LiDAR Observation

Figure 4 shows the selected map and the simulated 2D LiDAR sensor. The robot pose determines the origin and orientation of the scan, while each ray is cast into the occupancy grid until it reaches an obstacle, exits the map, or reaches the maximum range. The resulting set of distances forms the observation used by the localization algorithm.

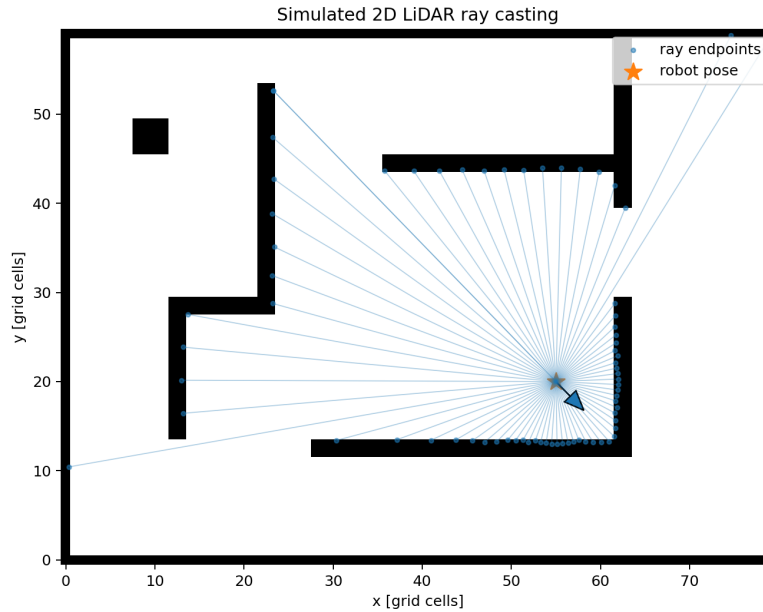


Figure 4: Simulated 2D LiDAR sensing in the `medium_room` map, shown in Figure 1c. The robot pose defines the origin of the scan, while the rays represent the range beams used to generate the observation.

A.1.2 Scan Comparison During Optimization

The evolutionary algorithm evaluates each candidate pose by comparing the observed scan with the scan simulated from that pose. Figure 5 shows this comparison at three stages of a representative run: the initial generation, an intermediate generation, and the final generation.

At the beginning of the optimization, the best candidate is usually far from the true pose, and its simulated scan differs significantly from the observed scan. As selection and variation operators act on the population, candidates that produce more compatible scans are favored. By the final generation, the simulated scan of the best individual is much closer to the observed one, indicating improved scan-map consistency.

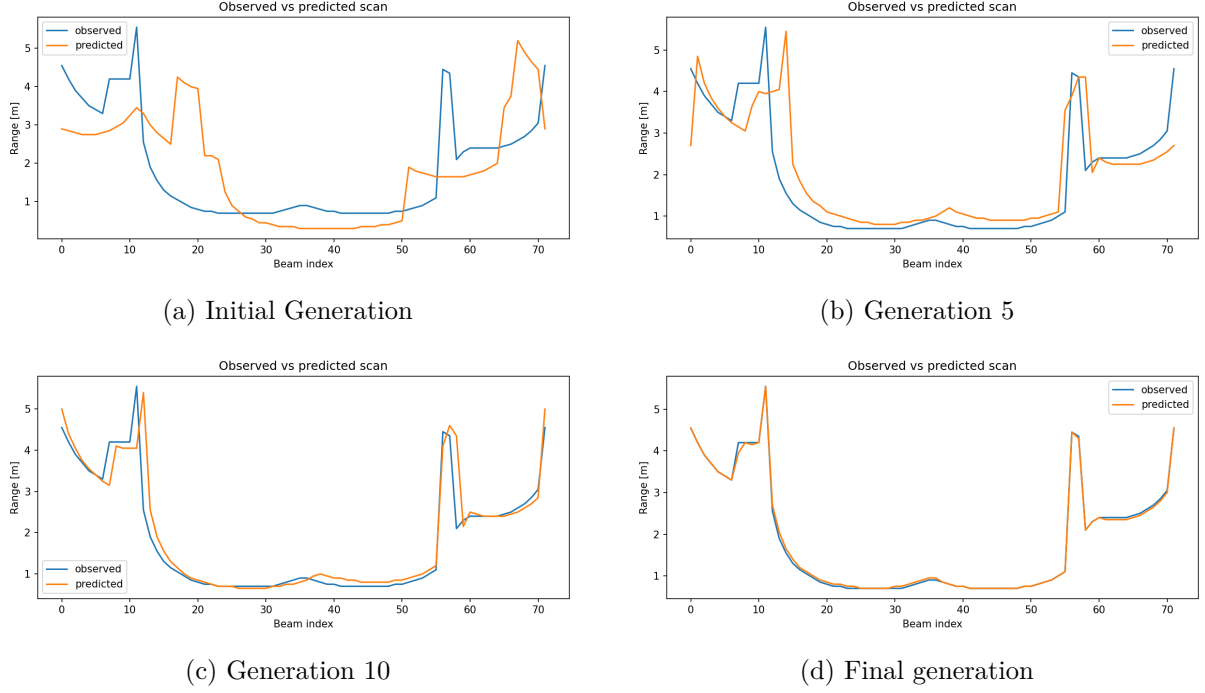
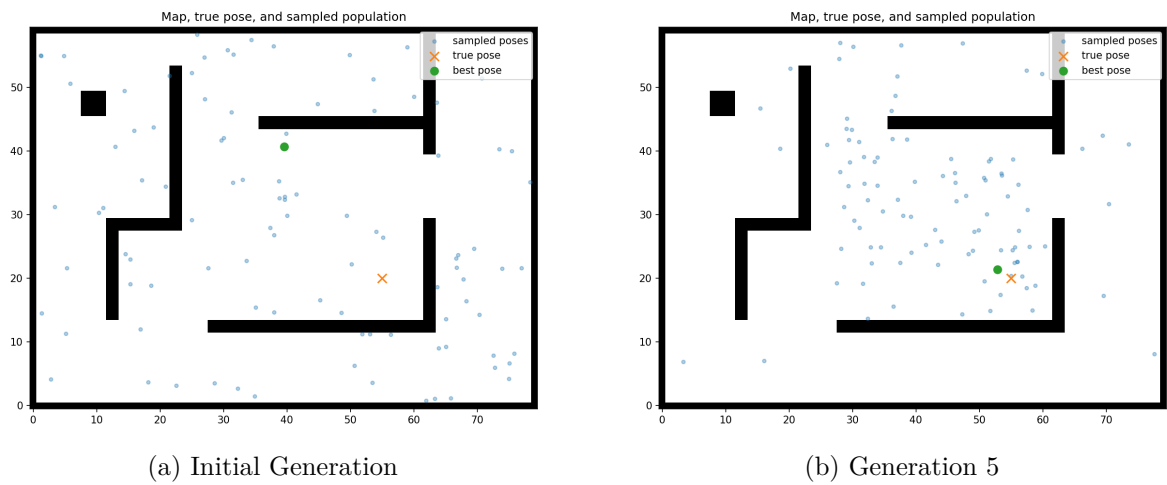


Figure 5: Scan comparison between the observed scan and the scan simulated from the best individual at different stages of the evolutionary process. The increasing overlap between the curves illustrates the improvement of scan consistency during optimization.

A.1.3 Population Convergence

Figure 6 shows the evolution of the population over multiple generations. At generation 0, individuals are spread across the valid pose space, reflecting the absence of prior information in the global localization setting. During the run, selection increases the representation of high-fitness regions, while crossover and mutation continue to generate new candidate poses around promising areas.

An animated version of the same convergence process is also provided as supplementary material in the file `population_convergence.gif`.



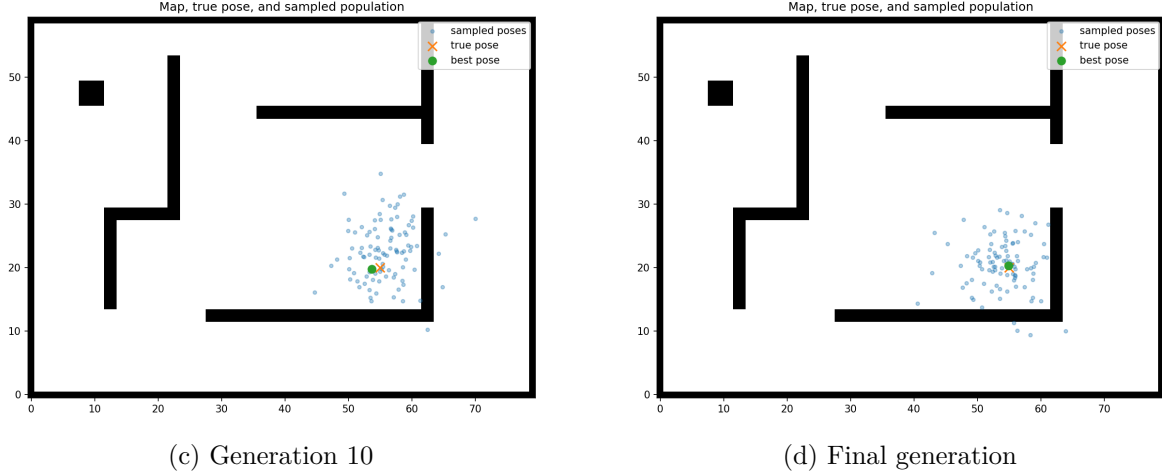


Figure 6: Population convergence during a representative EA localization run. The snapshots show the transition from broad exploration of the pose space to concentration around high-fitness pose hypotheses.

A.2 CSV Summary Files

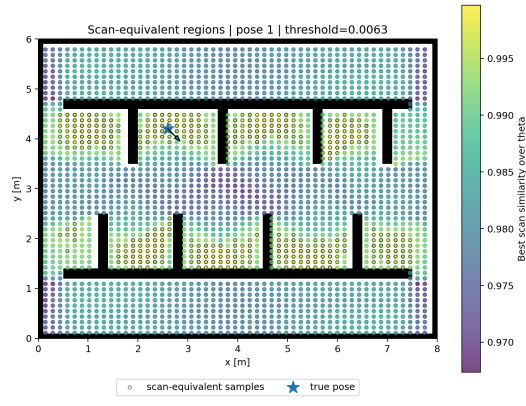
The following CSV files contain the most informative aggregated outputs used to build the quantitative results for this project, some of these tables are reported in Section 5.

- `experiment_1_summary_by_method`: summary of the perception ablation experiment. The results are grouped by fitness mode, reported in Table 5.
- `experiment_1_summary_by_map_name_method`: summary of the perception ablation experiment. The results are grouped by fitness mode and map name.
- `experiment_2_summary_by_maps_name.csv`: summary of the benchmark across all maps, reported in Table 6.
- `experiment_3_summary_by_config_id_map_name`: summary of the beam sensitivity experiment, grouped by map name and number of beams, reported in Table 7.
- `experiment_4_summary_by_config_id`: summary of the EA parameter tuning experiment, grouped by parameter change, reported in Table 8.
- `experiment_4_summary_by_config_id_map_name`: summary of the EA parameter tuning experiment, grouped by parameter change and map name.
- `experiment_5_symmetry_summary_by_map_name`: summary of the ambiguity and symmetry analysis, grouped by map name and represented in Table 9.
- `experiment_5_symmetry_summary_by_failure_type`: summary of the ambiguity and symmetry analysis, grouped by failure type and represented in Table 10.

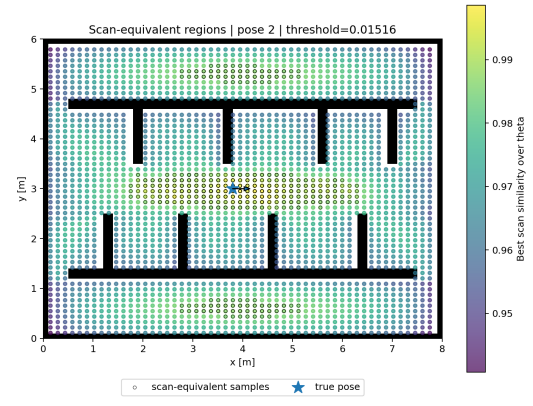
The project folder contains more than these tables, however some were just used as building blocks, other to compare information and other were just uninformative or redundant.

A.3 Additional Symmetry Maps

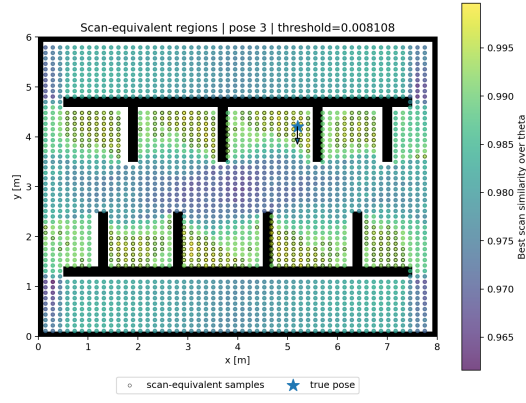
This appendix reports the remaining scan-equivalence maps not shown in the main text. Corridor pose 0 and complex room pose 2 are discussed in Section 5.5 and shown in Figures 2a and 2b; the figures below cover the remaining poses



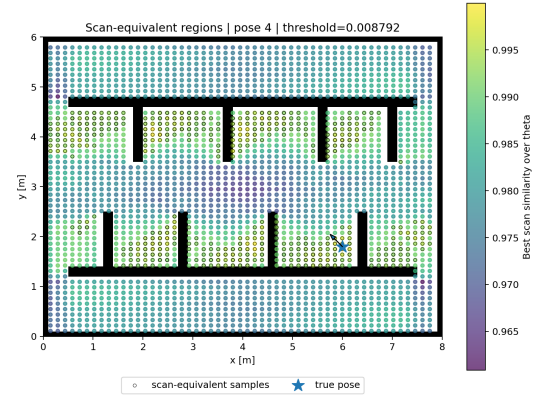
(a) corridor, pose 1



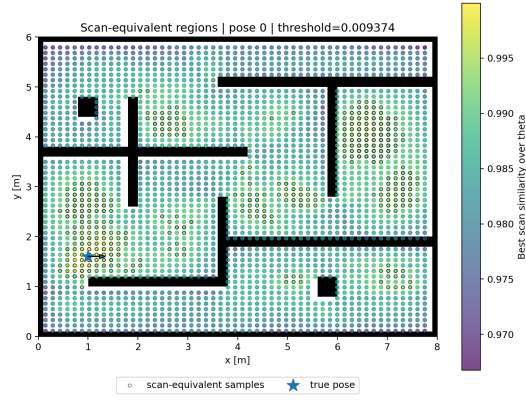
(b) corridor, pose 2



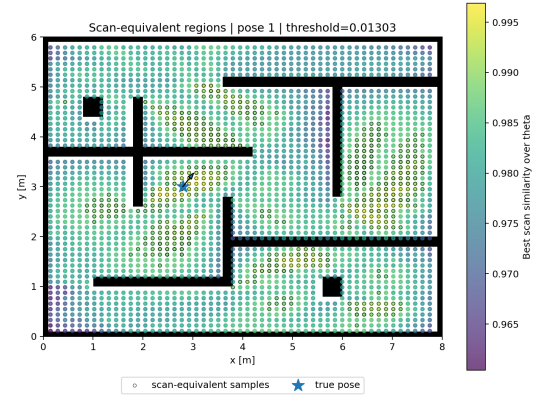
(c) corridor, pose 3



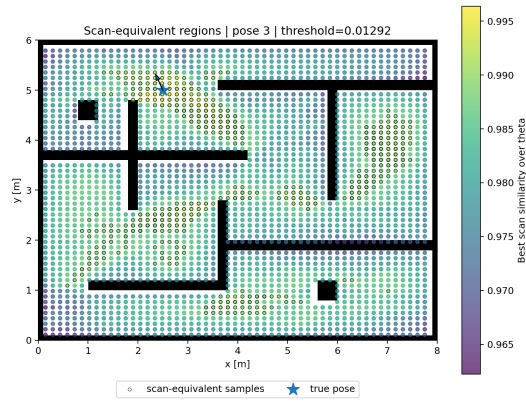
(d) corridor, pose 5



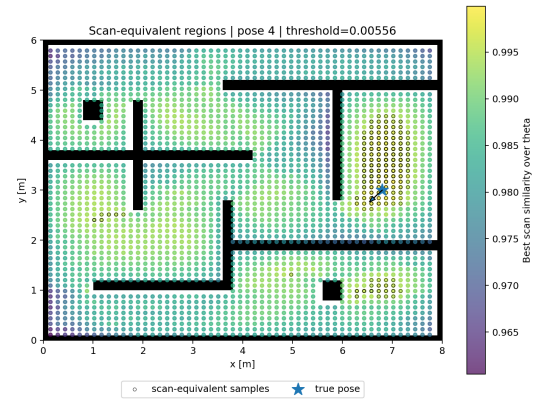
(e) complex_room, pose 0



(f) complex_room, pose 1



(g) complex_room, pose 3



(h) complex_room, pose 4

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